
$\mathbb{Z}_2$ Bosonic Spin Liquids on a Square Lattice

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ABSTRACT

Quantum spin liquids (QSLs) are exotic phases of matter that defy conventional magnetic ordering even at zero temperature due to strong quantum fluctuations. Among various classes of QSLs, the $\mathbb{Z}_2$ spin liquids have garnered significant attention for their potential role in topological quantum computation and emergent gauge structures. In this project, I investigated $\mathbb{Z}_2$ quantum spin liquids on a square lattice using the Schwinger boson formalism applied to the J_1 – J_2 Heisenberg model. Starting from the Schwinger boson representation of spin operators, I introduced pairing and hopping operators that facilitate the reformulation of the Hamiltonian. A Hubbard–Stratonovich transformation is then employed, followed by a mean-field approximation, yielding a tractable mean-field Hamiltonian. To identify possible spin liquid states, I used the results of Projective Symmetry Group (PSG) analysis, which constrains the mean-field ansätze compatible with lattice symmetries. This leads to the classification of two distinct $\mathbb{Z}_2$ spin liquid phases: the zero-flux and π -flux states, determined by specific conditions on the pairing and hopping amplitudes. The Hamiltonian is subsequently transformed to momentum space, but due to the nature of bosonic excitations in $\mathbb{Z}_2$ spin liquids, it remains non-diagonal. Hence, I performed Bogoliubov transformation to diagonalize the system and extract the dispersion relations, along with the Bogoliubov coefficients and corresponding self-consistency equations. Finally, I analyzed the parameter regimes for which self-consistent solutions exist.

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1 Introduction

Quantum spin liquids (QSLs) are ground states of spin systems in two or higher spatial dimensions that do not exhibit conventional magnetic ordering, even at zero temperature. This absence of long-range order is not a result of thermal fluctuations but rather arises from strong quantum fluctuations that persist in the ground state and prevent spontaneous symmetry breaking. These phases often emerge in frustrated spin- $\frac{1}{2}$ systems with a macroscopically degenerate classical ground state, where quantum interference among the degenerate configurations plays a crucial role in stabilizing the disordered phases. The highly entangled nature of QSLs leads to fundamentally new physical phenomena, such as fractionalized excitations, non-local correlations, and emergent gauge fields.

Quantum spin liquids are notable for exhibiting long-range quantum entanglement, which serves as a defining feature of their phase structure. This intrinsic entanglement underlies several exotic characteristics, including topological order, mutual statistics of emergent excitations, and gapless or gapped spinon spectra. As such, QSLs serve as key examples of topological phases of matter and are actively studied in both theoretical and experimental condensed matter physics.

Theoretical approaches to studying QSLs often rely on Parton constructions and effective gauge theories, which allow for the systematic classification and analysis of possible spin-liquid phases. Among these, $\mathbb{Z}_2$ spin liquids represent a particularly rich class of states, characterized by discrete gauge structures and topological ground state degeneracy. This work focuses on analyzing such $\mathbb{Z}_2$ spin liquid phases on the square lattice using the Schwinger boson formalism, applied to the frustrated J_1 - J_2 Heisenberg model.

$$\hat{\mathcal{H}} = J_1 \sum_{\langle i,j \rangle} \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j + J_2 \sum_{\langle\langle i,j \rangle\rangle} \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j \quad (1)$$

where $\hat{\vec{S}}_i$ is the spin operator at lattice site i , typically taken to be a spin- $\frac{1}{2}$ operator satisfying the standard $\text{SU}(2)$ algebra. The notation $\langle i,j \rangle$ denotes a summation over nearest-neighbor (NN) pairs of sites, and the J_1 term represents the nearest-neighbor Heisenberg exchange interaction. Similarly, $\langle\langle i,j \rangle\rangle$ denotes a summation over next-nearest-neighbor (NNN) pairs of sites, and the J_2 term accounts for the next-nearest-neighbor exchange interaction.

For antiferromagnetic interactions ($J_1 > 0$ and $J_2 > 0$), the presence of both NN and NNN terms introduces geometric frustration into the system. This frustration arises due to the competition between minimizing the energy of the NN and NNN interactions simultaneously. As a result, the classical magnetic order may be destabilized, giving rise to quantum-disordered ground states such as quantum spin liquids. The J_1 - J_2 model thus serves as a prototypical framework for exploring frustrated magnetism and nontrivial quantum phases.

2 Obtaining the Mean-field Hamiltonian

I have employed the Parton construction method to study the quantum spin liquids in frustrated spin systems, through the Schwinger boson mean-field theory which maps spin operators to bosonic degrees of freedom. In the Schwinger boson formalism, the spin operators at each site ‘ i ’ are expressed in terms of bosonic creation and annihilation operators as

$$\hat{\vec{S}}_i = \frac{1}{2} \sum_{\alpha, \beta} \hat{b}_{i\alpha}^\dagger \vec{\sigma}_{\alpha\beta} \hat{b}_{i\beta} \quad (2)$$

where $\hat{b}_{i\alpha}^\dagger$ and $\hat{b}_{i\beta}$ are bosonic creation and annihilation operators for spin states $\alpha, \beta \in \{1 (\uparrow \text{ type}), 2 (\downarrow \text{ type})\}$ at site i , and $\vec{\sigma} = (\sigma^x, \sigma^y, \sigma^z)$ is the vector of Pauli matrices. From here onward, Einstein’s summation convention is employed whenever applicable. These bosons obey the canonical commutation relations:

$$[\hat{b}_{i\alpha}, \hat{b}_{j\beta}^\dagger] = \delta_{ij} \delta_{\alpha\beta}, \quad [\hat{b}_{i\alpha}, \hat{b}_{j\beta}] = [\hat{b}_{i\alpha}^\dagger, \hat{b}_{j\beta}^\dagger] = 0.$$

This representation maps the spin- $\frac{1}{2}$ operator algebra into a bosonic Hilbert space. To ensure that the bosonic representation correctly describes a spin- $\frac{1}{2}$ system, a local constraint must be imposed at each site:

$$\hat{n}_i = \hat{b}_{i\alpha}^\dagger \hat{b}_{i\alpha} = 2S = 1, \quad (3)$$

for $S = \frac{1}{2}$. This constraint enforces that only one boson occupies each site, corresponding to the physical Hilbert space of a spin- $\frac{1}{2}$ particle.

The Heisenberg interaction $\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j$ can be written in terms of the Schwinger bosons as,

$$\hat{\vec{S}}_i \cdot \hat{\vec{S}}_j = \frac{1}{4} \hat{b}_{i\alpha}^\dagger \hat{b}_{i\beta} \hat{b}_{j\mu}^\dagger \hat{b}_{j\nu} (\vec{\sigma}_{\alpha\beta} \cdot \vec{\sigma}_{\mu\nu}) \quad (4)$$

$$= \frac{1}{2} \hat{b}_{i\alpha}^\dagger \hat{b}_{i\beta} \hat{b}_{j\beta}^\dagger \hat{b}_{j\alpha} - \frac{1}{4} \hat{b}_{i\alpha}^\dagger \hat{b}_{i\alpha} \hat{b}_{j\beta}^\dagger \hat{b}_{j\beta} \quad (5)$$

Where I have used the Pauli matrices identity $\vec{\sigma}_{\alpha\beta} \cdot \vec{\sigma}_{\mu\nu} = 2\delta_{\alpha\nu}\delta_{\beta\mu} - \delta_{\alpha\beta}\delta_{\mu\nu}$. Using the definitions of bond-pairing operator ($\hat{A}_{ij}$) and the bond-hopping operator ($\hat{B}_{ij}$):

$$\hat{A}_{ij} = \frac{1}{2} \epsilon_{\alpha\beta} \hat{b}_{i\alpha} \hat{b}_{j\beta} \quad (6)$$

$$\hat{B}_{ij} = \frac{1}{2} \hat{b}_{i\alpha}^\dagger \hat{b}_{j\alpha} \quad (7)$$

$$\implies \hat{\vec{S}}_i \cdot \hat{\vec{S}}_j = \hat{B}_{ij}^\dagger \hat{B}_{ij} - \hat{A}_{ij}^\dagger \hat{A}_{ij} \quad (8)$$

In writing the last equation, I have used normal ordering to remove nonphysical terms.

Thus, the J_1 - J_2 Heisenberg Hamiltonian can be written as,

$$\hat{\mathcal{H}} = J_1 \sum_{\langle i,j \rangle} \left(\hat{B}_{ij}^\dagger \hat{B}_{ij} - \hat{A}_{ij}^\dagger \hat{A}_{ij} \right) + J_2 \sum_{\langle\langle i,j \rangle\rangle} \left(\hat{B}_{ij}^\dagger \hat{B}_{ij} - \hat{A}_{ij}^\dagger \hat{A}_{ij} \right) \quad (9)$$

The next task is to perform the mean-field decoupling of these terms using the Hubbard-Stratonovich transformation. For this, I shall write the Heisenberg Hamiltonian in terms of the bond operators as

$$\hat{\mathcal{H}} = \sum_{i,j} J_{ij} \left(\hat{B}_{ij}^\dagger \hat{B}_{ij} - \hat{A}_{ij}^\dagger \hat{A}_{ij} \right) \quad (10)$$

Where J_{ij} is J_1 for the nearest sites and is J_2 for the next nearest sites. To proceed, I introduce complex auxiliary fields A_{ij} and B_{ij} , and the decoupling of the pairing term and the hopping term is achieved as follows:

$$e^{J_{ij} \hat{A}_{ij}^\dagger \hat{A}_{ij}} = \int [\mathcal{D}A_{ij}] \exp \left(-\frac{|A_{ij}|^2}{J_{ij}} + A_{ij}^* \hat{A}_{ij} + A_{ij} \hat{A}_{ij}^\dagger \right) \quad (11)$$

$$e^{-J_{ij} \hat{B}_{ij}^\dagger \hat{B}_{ij}} = \int [\mathcal{D}B_{ij}] \exp \left(-\frac{|B_{ij}|^2}{J_{ij}} + B_{ij}^* \hat{B}_{ij} + B_{ij} \hat{B}_{ij}^\dagger \right) \quad (12)$$

where $A_{ij} = -A_{ji}$ and $B_{ij} = B_{ji}^*$. To enforce the bosonic number constraint $\langle \hat{n}_i \rangle = \kappa = 2S$ on each lattice site, a real-valued Lagrange multiplier μ_i is introduced. Within the mean-field approximation, this multiplier is taken to be spatially uniform, i.e., independent of the site index i . In the mean-field approximation, the auxiliary fields are replaced with their saddle-point values leading to the self-consistency equations, and the Lagrange multiplier μ is treated as a variational parameter enforcing the average boson constraint.

$$A_{ij} = J_{ij} \langle \hat{A}_{ij} \rangle, \quad B_{ij} = J_{ij} \langle \hat{B}_{ij} \rangle, \quad \kappa = \langle \hat{n}_i \rangle \quad (13)$$

Here, the $\langle \dots \rangle$ represents the expectation value with respect to the ground state in which the Hamiltonian is diagonal. Substituting these into the Hamiltonian yields the mean-field form:

$$\hat{\mathcal{H}}_{\text{MF}} = \sum_{i,j} \left(B_{ij}^* \hat{B}_{ij} + B_{ij} \hat{B}_{ij}^\dagger - \frac{|B_{ij}|^2}{J_{ij}} - A_{ij}^* \hat{A}_{ij} - A_{ij} \hat{A}_{ij}^\dagger + \frac{|A_{ij}|^2}{J_{ij}} \right) - \sum_i \mu (\hat{n}_i - \kappa) \quad (14)$$

The Schwinger boson mean-field theory exhibits an emergent local $U(1)$ gauge symmetry, where the bosonic operators transform as $b_{j\alpha} \rightarrow e^{i\phi(j)} b_{j\alpha}$ without affecting any physical observables. Under this transformation, the mean-field amplitudes transform as $A_{ij} \rightarrow e^{i(\phi(i) + \phi(j))} A_{ij}$ and $B_{ij} \rightarrow e^{i(-\phi(i) + \phi(j))} B_{ij}$, reflecting the gauge redundancy of the parton construction. Because physical observables like spin correlations are gauge-invariant, two different mean-field ansätze related by such transformations describe the same physical quantum state. This leads to the classification of spin liquids using the Projective Symmetry Group (PSG), which captures symmetry realizations modulo gauge equivalences.

3 Zero flux $\mathbb{Z}_2$ QSL

The $\mathbb{Z}_2[0, 0]$ spin liquid state corresponds to a mean-field ansatz with zero gauge-invariant flux through all elementary plaquettes of the lattice. This is achieved by appropriate choices of the pairing amplitudes A_{ij} and hopping amplitudes B_{ij} between lattice sites. These mean-field parameters are given as follows.

$$\begin{aligned} A_{(x,y),(x+1,y)} &= -A_{(x,y),(x,y+1)} = A_1 \\ B_{(x,y),(x+1,y)} &= -B_{(x,y),(x,y+1)} = B_1 \\ B_{(x,y),(x+1,y+1)} &= -B_{(x,y),(x-1,y+1)} = B_2 \end{aligned} \quad (15)$$

Specifically, for this ansatz, A_1 (nearest-neighbor singlet pairing) and B_2 (next-nearest-neighbor hopping) are chosen to be real, while B_1 (nearest-neighbor hopping) is purely imaginary. This model has no next nearest-neighbor pairing ($A_2 = 0$). The phases of these amplitudes are chosen carefully so that the gauge-invariant flux vanishes around each elementary plaquette, preserving the zero-flux condition. Using this ansatz in the mean field Hamiltonian (14),

$$\begin{aligned} \hat{\mathcal{H}}_{\mathbb{Z}_2[0,0]} &= \sum_{\langle i,j \rangle} \left(B_{1ij} \hat{B}_{ij}^\dagger - B_{1ij} \hat{B}_{ij} + \frac{B_1^2}{J_1} - A_{1ij} \hat{A}_{ij}^\dagger - A_{1ij} \hat{A}_{ij} + \frac{A_1^2}{J_1} \right) \\ &\quad + \sum_{\langle\langle i,j \rangle\rangle} \left(B_{2ij} \hat{B}_{ij}^\dagger - B_{2ij} \hat{B}_{ij} + \frac{B_2^2}{J_2} \right) - \sum_i \mu(\hat{n}_i - \kappa) \end{aligned} \quad (16)$$

where A_{1ij} , B_{1ij} , and B_{2ij} are given by (15). The summation is expanded as

$$\sum_{\langle i,j \rangle} = \sum_i \sum_{j \in nn(i)} = \sum_{(x_i, y_i)} \sum_{j \in \{(x_i+1, y_i), (x_i, y_i+1)\}} \quad (17)$$

To get the dispersion relation, going to Fourier space is necessary. The discrete Fourier transform for the bosonic operators is given by,

$$\hat{b}_{i\alpha} = \frac{1}{\sqrt{N}} \sum_{\vec{k}} e^{-i\vec{k} \cdot \vec{r}_i} \hat{b}_{\vec{k}\alpha} \quad \text{and} \quad \hat{b}_{i\alpha}^\dagger = \frac{1}{\sqrt{N}} \sum_{\vec{k}} e^{i\vec{k} \cdot \vec{r}_i} \hat{b}_{\vec{k}\alpha}^\dagger \quad (18)$$

After performing the discrete Fourier transform on the $\mathbb{Z}_2$ zero-flux Hamiltonian, we get,

$$\begin{aligned} \hat{\mathcal{H}}_{\mathbb{Z}_2[0,0]} &= iB_1 \sum_{\vec{k}} [\sin(k_x) - \sin(k_y)] \hat{b}_{\vec{k}\alpha}^\dagger \hat{b}_{\vec{k}\alpha} - 2B_2 \sum_{\vec{k}} \sin(k_x) \sin(k_y) \hat{b}_{\vec{k}\alpha}^\dagger \hat{b}_{\vec{k}\alpha} \\ &\quad - \frac{A_1 \epsilon_{\alpha\beta}}{2} \sum_{\vec{k}} \left[(e^{ik_x} - e^{iky}) \hat{b}_{-\vec{k}\alpha}^\dagger \hat{b}_{\vec{k}\beta}^\dagger + (e^{-ik_x} - e^{-iky}) \hat{b}_{-\vec{k}\alpha} \hat{b}_{\vec{k}\beta} \right] \\ &\quad - \mu \sum_{\vec{k}} \hat{b}_{\vec{k}\alpha}^\dagger \hat{b}_{\vec{k}\alpha} + N \left[\mu\kappa + 2 \left(\frac{A_1^2 + B_1^2}{J_1} \right) - \frac{2B_2^2}{J_2} \right] \end{aligned} \quad (19)$$

To simplify this expression, we expand the summation over spin indices α and β and rewrite the Hamiltonian in Nambu spinor form:

$$\hat{\mathcal{H}}_{\mathbb{Z}_2[0,0]} = \sum_{\vec{k}} \Psi_{\vec{k}}^\dagger H_{\vec{k}} \Psi_{\vec{k}} + N \left[\mu(\kappa + 1) + 2 \left(\frac{A_1^2 + B_1^2}{J_1} \right) - \frac{2B_2^2}{J_2} \right] \quad (20)$$

where

$$\Psi_{\vec{k}} = \begin{bmatrix} \hat{b}_{\vec{k}1} \\ \hat{b}_{-\vec{k}2}^\dagger \end{bmatrix}, \quad H_{\vec{k}} = \begin{bmatrix} iB_1 f_- - 2B_2 f_1 - \mu & iA_1 f_- \\ -iA_1 f_- & iB_1 f_- - 2B_2 f_1 - \mu \end{bmatrix} \quad (21)$$

with the shorthand functions defined as:

$$f_1 = \sin(k_x) \sin(k_y), \quad f_{\pm} = \sin(k_x) \pm \sin(k_y)$$

To obtain the bosonic dispersion relation, the matrix $H_{\vec{k}}$ must be diagonalized. This is done through a bosonic Bogoliubov transformation, which relates the original Schwinger boson operators $(\hat{b}_{\vec{k}1}, \hat{b}_{\vec{k}2}, \hat{b}_{\vec{k}1}^\dagger, \hat{b}_{\vec{k}2}^\dagger)$ to new quasi-particle bosonic operators $(\hat{a}_{\vec{k}1}, \hat{a}_{\vec{k}2}, \hat{a}_{\vec{k}1}^\dagger, \hat{a}_{\vec{k}2}^\dagger)$ via a symplectic matrix belonging to the group $\text{Sp}(2, \mathbb{C})$:

$$\mathbf{U} = \begin{bmatrix} u & v \\ v^* & u^* \end{bmatrix} \implies \Psi_{\vec{k}} = \mathbf{U} \Phi_{\vec{k}} \quad \text{where} \quad \Phi_{\vec{k}} = \begin{bmatrix} \hat{a}_{\vec{k}1} \\ \hat{a}_{-\vec{k}2}^\dagger \end{bmatrix} \quad (22)$$

To preserve the bosonic commutation relations, the matrix elements of the Bogoliubov transformation must satisfy the condition $|u|^2 - |v|^2 = 1$. Under this transformation, the Hamiltonian becomes diagonal:

$$\Psi_{\vec{k}}^\dagger H_{\vec{k}} \Psi_{\vec{k}} = \Phi_{\vec{k}}^\dagger \mathbf{U}^\dagger H_{\vec{k}} \mathbf{U} \Phi_{\vec{k}} = \Phi_{\vec{k}}^\dagger \Omega_{\vec{k}} \Phi_{\vec{k}},$$

where $\Omega_{\vec{k}}$ is a diagonal matrix whose entries represent the energy dispersion of the quasi-particles. The elements of the Bogoliubov transformation matrix $\mathbf{U}$ are given by:

$$u = \frac{\chi}{\sqrt{\chi^2 - 1}}, \quad v = \frac{-i}{\sqrt{\chi^2 - 1}} \quad \text{with} \quad \chi = \frac{-(2B_2 f_1 + \mu) + \sqrt{(2B_2 f_1 + \mu)^2 - (A_1 f_-)^2}}{A_1 f_-} \quad (23)$$

Thus, the diagonalized Hamiltonian is expressed through the matrix $\Omega_{\vec{k}}$ as:

$$\Omega_{\vec{k}} = \begin{bmatrix} \omega_+ & 0 \\ 0 & \omega_- \end{bmatrix}, \quad (24)$$

with the dispersion relations $\omega_{\pm}$ as:

$$\omega_{\pm} = \sqrt{(2B_2 f_1 + \mu)^2 - (A_1 f_-)^2} \pm \text{Im}(B_1) f_-. \quad (25)$$

The next task is to determine the values of the mean-field parameters A_1 , B_1 , and B_2 using the self-consistency equations (13). The expectation value of the pairing operator is:

$$\langle A_{ij} \rangle = \frac{1}{2} \langle \epsilon_{\alpha\beta} \hat{b}_{i\alpha} \hat{b}_{j\beta} \rangle = \frac{1}{2N} \sum_{\vec{k}, \vec{q}} \left[e^{-i(\vec{k} \cdot \vec{r}_i - \vec{q} \cdot \vec{r}_j)} \langle \hat{b}_{\vec{k}1} \hat{b}_{-\vec{q}2} \rangle - e^{i(\vec{k} \cdot \vec{r}_i - \vec{q} \cdot \vec{r}_j)} \langle \hat{b}_{-\vec{k}2} \hat{b}_{\vec{q}1} \rangle \right] \quad (26)$$

Substituting the Schwinger boson operators using the Bogoliubov matrix we get,

$$\langle \hat{b}_{\vec{k}1} \hat{b}_{-\vec{q}2} \rangle = \langle \hat{b}_{-\vec{k}2} \hat{b}_{\vec{q}1} \rangle = u_{\vec{k}} v_{\vec{q}} \delta_{\vec{k}\vec{q}} \quad (27)$$

$$\Rightarrow \langle A_{ij} \rangle = -\frac{i}{2N} \sum_{\vec{k}} \left[e^{-i\vec{k} \cdot (\vec{r}_i - \vec{r}_j)} - e^{i\vec{k} \cdot (\vec{r}_i - \vec{r}_j)} \right] \left(\frac{\chi}{\chi^2 - 1} \right) \quad (28)$$

Using $A_1 = J_{ij} \langle \hat{A}_{ij} \rangle$ where $j \in \text{nn}(i)$, and employing the zero-flux ansatz (15), we get:

$$\frac{A_1}{J_1} = \frac{\langle A_{(x_i, y_i), (x_i+1, y_i)} \rangle - \langle A_{(x_i, y_i), (x_i, y_i+1)} \rangle}{2}, \quad (29)$$

$$\Rightarrow \frac{2A_1}{J_1} = \frac{i}{N} \sum_{\vec{k}} [\sin(k_x) - \sin(k_y)] \left(\frac{\chi}{\chi^2 - 1} \right). \quad (30)$$

Now, replacing the summation by an integral $\frac{1}{N} \sum_{\vec{k}} \rightarrow \int_{-\pi}^{\pi} \frac{d^2 k}{(2\pi)^2}$, and performing similar computations for the remaining mean field parameters B_1 , B_2 , and the particle number constraint $\langle \hat{n}_i \rangle$, we arrive at the following self-consistency equations:

$$\begin{aligned} \frac{4A_1}{J_1} &= \int_{-\pi}^{\pi} \frac{d^2 k}{(2\pi)^2} \frac{A_1 f_-^2}{\sqrt{(2B_2 f_1 + \mu)^2 - (A_1 f_-)^2}} \\ \frac{4B_1}{J_1} &= \int_{-\pi}^{\pi} \frac{d^2 k}{(2\pi)^2} \frac{[\cos(k_x) - \cos(k_y)](2B_2 f_1 + \mu)}{\sqrt{(2B_2 f_1 + \mu)^2 - (A_1 f_-)^2}} \\ \frac{2B_2}{J_2} &= \int_{-\pi}^{\pi} \frac{d^2 k}{(2\pi)^2} \frac{f_1 (2B_2 f_1 + \mu)}{\sqrt{(2B_2 f_1 + \mu)^2 - (A_1 f_-)^2}} \\ \kappa + 1 &= \int_{-\pi}^{\pi} \frac{d^2 k}{(2\pi)^2} \frac{-(2B_2 f_1 + \mu)}{\sqrt{(2B_2 f_1 + \mu)^2 - (A_1 f_-)^2}} \end{aligned} \quad (31)$$

A closer inspection of the self-consistency equations reveals that the parameter B_1 does not appear in any of them and is zero. Among the remaining equations, only the third one—corresponding to B_2 is actually self-consistent. It must be solved iteratively for B_2 , using initial guess values for A_1 , B_2 , and μ . Once the value of B_2 converges, it should be substituted into the first and fourth equations, which can then be treated as root-finding problems for A_1 and μ , respectively. This process is carried out for fixed values of J_1 , J_2 , and κ .

4 π flux $\mathbb{Z}_2$ QSL

The π -flux $\mathbb{Z}_2[\pi, 0]$ spin liquid state is characterized by a mean-field ansatz that enforces a π gauge-invariant flux through every elementary plaquette of the lattice, i.e., a phase of π is accumulated by a spinon hopping around a plaquette. This flux pattern is realized through specific choices of the singlet pairing amplitudes A_{ij} and the hopping amplitudes B_{ij} between neighboring lattice sites. The structure of the ansatz is encoded in the following pattern of the pairing amplitudes:

$$\begin{aligned} (-1)^y A_{(x,y),(x+1,y)} &= -A_{(x,y),(x,y+1)} = A_1 \\ (-1)^y A_{(x,y),(x+1,y+1)} &= (-1)^{y+1} A_{(x,y),(x-1,y+1)} = A_2 \end{aligned} \quad (32)$$

In this ansatz, the parameters A_1 and A_2 , corresponding to nearest-neighbor and next-nearest-neighbor singlet pairings respectively, are taken to be real. The hopping amplitudes B_1 and B_2 are set to zero, reflecting the absence of spinon hopping in this particular spin liquid state.

The structure of this ansatz distinguishes between lattice sites with even and odd values of the y -coordinate, effectively breaking the original translational symmetry along the y -direction. As a result, the periodicity along y is doubled, and the elementary unit cell becomes rectangular with dimensions 1×2 , spanning one site along the x -axis and two sites along the y -axis. Each unit cell, labeled by coordinates $(\tilde{x}, \tilde{y})$, contains two inequivalent lattice sites denoted by ‘ u ’ and ‘ v ’, located at $(x, 2y)$ and $(x, 2y + 1)$, respectively.

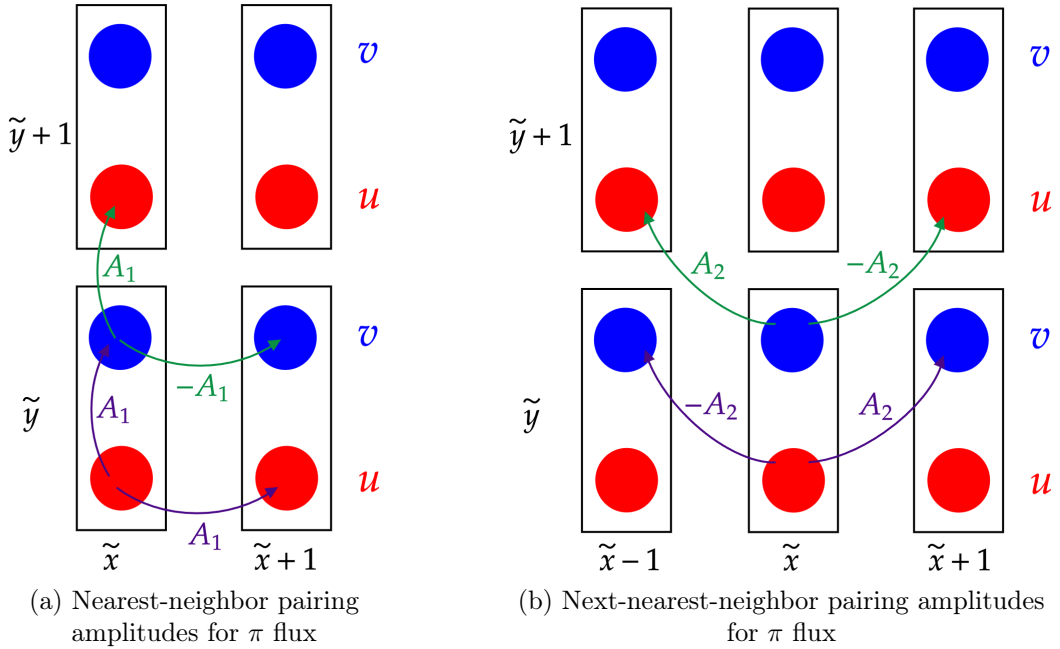


Figure 1: Schematic representation of the enlarged unit cell containing two types of sites: ‘ u ’ (even y) and ‘ v ’ (odd y). Arrows indicate the sign structure of the nearest- and next-nearest-neighbor pairing amplitudes in the π -flux ansatz.

The Hamiltonian for the $\mathbb{Z}_2[\pi, 0]$ quantum spin liquid in the Schwinger boson mean-field formulation is given by:

$$\hat{\mathcal{H}}_{\mathbb{Z}_2[\pi, 0]} = \sum_{\langle i, j \rangle} \left[\frac{-A_{1ij} \epsilon_{\alpha\beta}}{2} \left(\hat{b}_{i\alpha} \hat{b}_{j\beta} + \hat{b}_{i\alpha}^\dagger \hat{b}_{j\beta}^\dagger \right) \right] + \sum_{\langle\langle i, j \rangle\rangle} \left[\frac{-A_{2ij} \epsilon_{\alpha\beta}}{2} \left(\hat{b}_{i\alpha} \hat{b}_{j\beta} + \hat{b}_{i\alpha}^\dagger \hat{b}_{j\beta}^\dagger \right) \right] - \sum_i \mu(\hat{n}_i - \kappa) + 2N \left(\frac{A_1^2}{J_1} + \frac{A_2^2}{J_2} \right) \quad (33)$$

The structure of the lattice is such that sites with even and odd y -coordinates are labeled differently, leading to a rectangular unit cell of size 1×2 . The explicit decomposition of the NN and NNN summations is given by:

$$\begin{aligned} \sum_{\langle i, j \rangle} &= \sum_{i \in u(x, 2y)} \sum_{j \in \{u(x+1, 2y), v(x, 2y)\}} + \sum_{i \in v(x, 2y)} \sum_{j \in \{v(x+1, 2y), u(x, 2y+2)\}} \\ \sum_{\langle\langle i, j \rangle\rangle} &= \sum_{i \in u(x, 2y)} \sum_{j \in \{v(x+1, 2y), v(x-1, 2y)\}} + \sum_{i \in v(x, 2y)} \sum_{j \in \{u(x+1, 2y+2), u(x-1, 2y+2)\}} \end{aligned} \quad (34)$$

To analyze the Hamiltonian in momentum space, we define the discrete Fourier transform for the boson operators in the enlarged unit cell as:

$$\begin{aligned} \hat{b}_{u(x, 2y)\alpha} &= \frac{1}{\sqrt{N}} \sum_{\vec{k}} e^{-i(k_x x + 2k_y y)} \hat{b}_{u\vec{k}\alpha} \\ \hat{b}_{v(x, 2y)\alpha} &= \frac{1}{\sqrt{N}} \sum_{\vec{k}} e^{-i(k_x x + 2k_y y)} \hat{b}_{v\vec{k}\alpha} \end{aligned} \quad (35)$$

The Fourier transform of the creation operators is obtained by taking the complex adjoint of the above equations. The orthogonality relation for the Fourier basis is:

$$\delta(\vec{k} - \vec{q}) = \sum_{(x, 2y)} \frac{e^{i(k_x - q_x)x}}{N_x} \frac{e^{2i(k_y - q_y)y}}{N_y} \quad (36)$$

Substituting these expressions into the real-space Hamiltonian gives the momentum-space form:

$$\begin{aligned} \hat{\mathcal{H}}_{\mathbb{Z}_2[\pi, 0]} &= \frac{A_1 \epsilon_{\alpha\beta}}{2} \sum_{\vec{k}} \left[-e^{ik_x} \left(\hat{b}_{u\vec{k}\alpha} \hat{b}_{u-\vec{k}\beta} + \hat{b}_{u-\vec{k}\alpha}^\dagger \hat{b}_{u\vec{k}\beta}^\dagger \right) + e^{ik_x} \left(\hat{b}_{v\vec{k}\alpha} \hat{b}_{v-\vec{k}\beta} + \hat{b}_{v-\vec{k}\alpha}^\dagger \hat{b}_{v\vec{k}\beta}^\dagger \right) \right. \\ &\quad \left. - \left(\hat{b}_{u\vec{k}\alpha} \hat{b}_{v-\vec{k}\beta} + \hat{b}_{u-\vec{k}\alpha}^\dagger \hat{b}_{v\vec{k}\beta}^\dagger \right) - e^{2ik_y} \left(\hat{b}_{v\vec{k}\alpha} \hat{b}_{u-\vec{k}\beta} + \hat{b}_{v-\vec{k}\alpha}^\dagger \hat{b}_{u\vec{k}\beta}^\dagger \right) \right] \\ &\quad + iA_2 \epsilon_{\alpha\beta} \sum_{\vec{k}} \sin(k_x) \left[\hat{b}_{u\vec{k}\alpha}^\dagger \hat{b}_{v-\vec{k}\beta}^\dagger - \hat{b}_{u\vec{k}\alpha} \hat{b}_{v-\vec{k}\beta} + e^{2ik_y} \left(\hat{b}_{v\vec{k}\alpha} \hat{b}_{u-\vec{k}\beta} + \hat{b}_{v-\vec{k}\alpha}^\dagger \hat{b}_{u\vec{k}\beta}^\dagger \right) \right] \\ &\quad - \mu \sum_{\vec{k}} \left[\hat{b}_{u\vec{k}\alpha}^\dagger \hat{b}_{u\vec{k}\alpha} + \hat{b}_{v\vec{k}\alpha}^\dagger \hat{b}_{v\vec{k}\alpha} \right] - N\mu\kappa + 2N \left(\frac{A_1^2}{J_1} + \frac{A_2^2}{J_2} \right) \end{aligned} \quad (37)$$

To simplify the Hamiltonian, we expand the summation over spin indices α and β , and express it in terms of Nambu spinor,

$$\hat{\mathcal{H}}_{\mathbb{Z}_2[\pi,0]} = \sum_{\vec{k}} \Psi_{\vec{k}}^\dagger D_{\vec{k}} \Psi_{\vec{k}} + N \left[\mu(\kappa + 1) + \frac{2A_1^2}{J_1} + \frac{2A_2^2}{J_2} \right] \quad (38)$$

where the Nambu spinor is defined as

$$\Psi_{\vec{k}} = \begin{bmatrix} \hat{b}_{u\vec{k}1} \\ \hat{b}_{v\vec{k}1} \\ \hat{b}_{u-\vec{k}2}^\dagger \\ \hat{b}_{v-\vec{k}2}^\dagger \end{bmatrix}, \quad \text{and} \quad D_{\vec{k}} = \begin{bmatrix} -\mu \mathbf{i}_2 & i\Delta_{\vec{k}} \\ -i\Delta_{\vec{k}} & -\mu \mathbf{i}_2 \end{bmatrix} \quad (39)$$

Here, $D_{\vec{k}}$ is a $2n \times 2n$ Hermitian, with $n = 2$, $\mathbf{i}_2$ is the 2×2 identity matrix, and $\Delta_{\vec{k}}$ is a 2×2 hermitian pairing matrix given by,

$$\Delta_{\vec{k}} = A_1 P_{\vec{k}}^{(1)} + A_2 P_{\vec{k}}^{(2)} \quad (40)$$

with

$$P_{\vec{k}}^{(1)} = \begin{bmatrix} \sin(k_x) & \frac{i}{2}(1 - e^{2ik_y}) \\ -\frac{i}{2}(1 - e^{-2ik_y}) & -\sin(k_x) \end{bmatrix}, \quad P_{\vec{k}}^{(2)} = \begin{bmatrix} 0 & \sin(k_x)(1 - e^{2ik_y}) \\ \sin(k_x)(1 - e^{-2ik_y}) & 0 \end{bmatrix} \quad (41)$$

To extract the dispersion relation, we diagonalize the quadratic Hamiltonian written in Nambu spinor form using the bosonic Bogoliubov–de Gennes (BdG) formalism. The Hamiltonian is encoded in the matrix $D_{\vec{k}}$, which is a $2n \times 2n$ Hermitian matrix ($n = 2$ in our case), and $\Psi_{\vec{k}}$ is the corresponding $2n$ -component Nambu spinor constructed from bosonic annihilation and creation operators. To preserve the bosonic commutation relations, the transformation matrix $\mathbf{U}_{\vec{k}}$ must satisfy the symplectic condition:

$$\mathbf{U}_{\vec{k}}^\dagger \Sigma_z \mathbf{U}_{\vec{k}} = \Sigma_z, \quad \text{where} \quad \Sigma_z = \begin{bmatrix} \mathbf{i}_n & 0 \\ 0 & -\mathbf{i}_n \end{bmatrix} \quad (42)$$

This condition ensures that the transformed operators obey the bosonic algebra. We then diagonalize the non-Hermitian matrix $\Sigma_z D_{\vec{k}}$. The eigenvalue problem to solve is:

$$\Sigma_z D_{\vec{k}} \mathbf{U}_{\vec{k}} = \mathbf{U}_{\vec{k}} \Omega_{\vec{k}} \quad (43)$$

The positive eigenvalues of $\Sigma_z D_{\vec{k}}$ correspond to the physical bosonic modes.

The diagonal form of the Hamiltonian is given by:

$$\hat{\mathcal{H}}_{\mathbb{Z}_2[\pi,0]} = \sum_{\vec{k}} \Phi_{\vec{k}}^\dagger \Omega_{\vec{k}} \Phi_{\vec{k}} + N \left[\mu(\kappa + 1) + \frac{2A_1^2}{J_1} + \frac{2A_2^2}{J_2} \right] \quad (44)$$

where $\Phi_{\vec{k}} = \mathbf{U}_{\vec{k}}^{-1} \Psi_{\vec{k}}$ is the Bogoliubov quasiparticle spinor, and $\Omega_{\vec{k}}$ is a diagonal matrix whose entries define the dispersion relation. The dispersion for the π -flux $\mathbb{Z}_2$ quantum spin liquid is:

$$\omega(\vec{k}) = \sqrt{\mu^2 - A_1^2(f_-^2 + 2f_1) - (4A_2f_1)^2} \quad (45)$$

The mean-field self-consistency equations are obtained by extremizing the dispersion relation. They are given by:

$$\begin{aligned} \frac{4}{J_1} &= \int_{-\pi}^{\pi} \frac{d^2k}{(2\pi)^2} \frac{(f_-^2 + 2f_1)}{\sqrt{\mu^2 - A_1^2(f_-^2 + 2f_1) - (4A_2f_1)^2}} \\ \frac{4}{J_2} &= \int_{-\pi}^{\pi} \frac{d^2k}{(2\pi)^2} \frac{(4f_1)^2}{\sqrt{\mu^2 - A_1^2(f_-^2 + 2f_1) - (4A_2f_1)^2}} \\ 1 + \kappa &= \int_{-\pi}^{\pi} \frac{d^2k}{(2\pi)^2} \frac{-\mu}{\sqrt{\mu^2 - A_1^2(f_-^2 + 2f_1) - (4A_2f_1)^2}} \end{aligned} \quad (46)$$

By solving these equations for the chosen values of J_1 , J_2 , and κ , we can determine the mean-field parameters A_1 , A_2 , and μ . The following algorithm describes the process for computing the Bogoliubov matrix after obtaining the mean-field parameters:

Algorithm: Compute Bosonic Bogoliubov Matrix $\mathbf{U}_{\vec{k}}$

- 1: For each point on the Brillouin zone, Compute $M_{\vec{k}} = \Sigma_z D_{\vec{k}}$
 - 2: Solve $M_{\vec{k}} \vec{v}_i = \omega_i(\vec{k}) \vec{v}_i$ for eigenvalues ω_i and eigenvectors $\vec{v}_i$
 - 3: Normalize each $\vec{v}_i$ such that $\vec{v}_i^\dagger \Sigma_z \vec{v}_i = \pm 1$
 - 4: Retain only the eigenvectors with positive-norm and positive-frequency
 - 5: Form $\mathbf{U}_{\vec{k}} = [\vec{v}_1, \vec{v}_2, \dots, \vec{v}_n, \vec{v}_1^*, \dots, \vec{v}_n^*]$
-

5 Numerical Analysis

In this section, a numerical framework is developed to solve the coupled self-consistency and root-finding equations of the $\mathbb{Z}_2[0,0]$ zero-flux mean-field model. The core of the method rests on three key integrals over the two-dimensional Brillouin zone,

$$A_1 = \frac{1}{(2\pi)^2} \iint_{\text{B.Z.}} \frac{J_1 A_1 [\sin(k_x) - \sin(k_y)]^2}{4 \sqrt{[\mu + 2B_2 \sin(k_x) \sin(k_y)]^2 - A_1^2 [\sin(k_x) - \sin(k_y)]^2}} dk_x dk_y \quad (47a)$$

$$B_2 = \frac{1}{(2\pi)^2} \iint_{\text{B.Z.}} \frac{J_2 \sin(k_x) \sin(k_y) [\mu + 2B_2 \sin(k_x) \sin(k_y)]}{2 \sqrt{[\mu + 2B_2 \sin(k_x) \sin(k_y)]^2 - A_1^2 [\sin(k_x) - \sin(k_y)]^2}} dk_x dk_y \quad (47b)$$

$$\kappa + 1 = \frac{1}{(2\pi)^2} \iint_{\text{B.Z.}} \frac{-[\mu + 2B_2 \sin(k_x) \sin(k_y)]}{\sqrt{[\mu + 2B_2 \sin(k_x) \sin(k_y)]^2 - A_1^2 [\sin(k_x) - \sin(k_y)]^2}} dk_x dk_y \quad (47c)$$

In what follows, the values of J_1 and J_2 are set to be 1 and seek the solutions for the parameters A_1 , B_2 , and μ which satisfy $\kappa > 0$.

5.1 Solution by fixed point iteration method for both A_1 and B_2

Here the first two equations (47a) and (47b) are solved iteratively, by the iteration scheme,

$$A_1^{n+1} = f_{A_1}(A_1^n) \quad \text{and} \quad B_2^{n+1} = f_{B_2}(B_2^n)$$

where each f is defined by the right-hand side of its respective equation. The initial guesses $A_1^{(0)}$, $B_2^{(0)}$ for a particular value of μ are chosen such that,

$$([\mu + 2B_2 \sin(k_x) \sin(k_y)]^2 - A_1^2 [\sin(k_x) - \sin(k_y)]^2) > 0, \forall \vec{k} \in \text{1st B.Z.}$$

Since, $\omega(\vec{k})$ has to be defined for all the $\vec{k}$ points in the 1st B.Z.. The iterations for A_1 and B_2 continues until $|A_1^{(n+1)} - A_1^{(n)}|$ and $|B_2^{(n+1)} - B_2^{(n)}|$ both fall below a prescribed tolerance. Thus, the values converge to fixed points $A_1^{(n)} \rightarrow A_1^*$ and $B_2^{(n)} \rightarrow B_2^*$ such that, the first two self-consistent equations are satisfied and then the corresponding value of $\kappa + 1$ is found.

Results

This procedure yields three classes of stationary points:

- The trivial paramagnet: $A_1^* = B_2^* = 0$, giving $\kappa + 1 = 1$.
- A spurious solution: $A_1^* = 0$, $B_2^* \neq 0$, with $\kappa + 1 = 0$.
- Complex pairing amplitude solution where A_1^* is purely imaginary, B_2 is purely real, and $-1 < \kappa + 1 < 1$ where positive (negative) μ produces negative (positive) $\kappa + 1$.

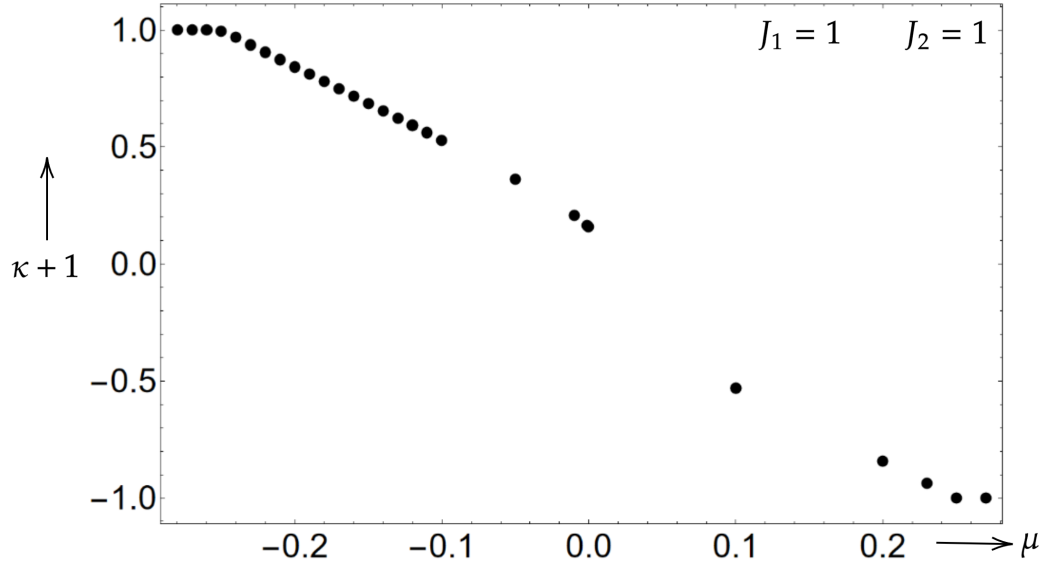


Figure 2: $\kappa + 1$ as a function of μ , got by fixed point iteration method for A_1 and B_2 .

A_1^*	B_2^*	μ	$\kappa + 1$
0.169776 i	-0.144823	-0.0001	0.159657
0.171958 i	-0.141817	-0.01	0.208067
0.176936 i	-0.126915	-0.05	0.363047
0.175108 i	-0.102690	-0.1	0.529455
0.173488 i	-0.097171	-0.11	0.561218
0.171372 i	-0.091439	-0.12	0.592713
0.168712 i	-0.085497	-0.13	0.624012
0.165449 i	-0.079350	-0.14	0.655176
0.161515 i	-0.073003	-0.15	0.686258
0.156823 i	-0.066461	-0.16	0.717304
0.151266 i	-0.059729	-0.17	0.748353
0.144704 i	-0.052814	-0.18	0.779440
0.136953 i	-0.045724	-0.19	0.810594
0.127753 i	-0.038466	-0.20	0.841840
0.116714 i	-0.031049	-0.21	0.873197
0.103199 i	-0.023482	-0.22	0.904681
0.085993 i	-0.015778	-0.23	0.936305
0.062038 i	-0.007949	-0.24	0.968066
0.022368 i	-0.001097	-0.25	0.995657
0.002403 i	-0.000011	-0.26	0.999957
5.912×10^{-5}	0.0000000	-0.27	1.000000
0.000000	0.0000000	-0.28	1.000000

Table 1: Fixed-point solutions (A_1^*, B_2^*) and resulting $\kappa + 1$ as μ varies. For $\mu < -0.28$, $A_1^* = 0$ and $B_2^* = 0$ leading to the trivial, paramagnetic phase without any pairing or hopping.

5.2 In the limit $J_2 = 0$, with $B_2 = 0$

Since the $\mathbb{Z}_2[0,0]$ flux model accepts a solution with no next-nearest neighbor hopping (i.e., with $J_2 = 0$), a physically acceptable solution with a real non-zero value for A^* can be expected here. This reduces the problem to two coupled integrals in the renormalized variables,

$$\xi = \frac{J_1}{-\mu}, \quad \gamma = \frac{A_1}{-\mu}.$$

The self-consistency conditions become

$$\frac{4}{\xi} = \frac{1}{(2\pi)^2} \iint_{\text{B.Z.}} \frac{[\sin(k_x) - \sin(k_y)]^2}{\sqrt{1 - \gamma^2 [\sin(k_x) - \sin(k_y)]^2}} dk_x dk_y \quad (48a)$$

$$\kappa + 1 = \frac{1}{(2\pi)^2} \iint_{\text{B.Z.}} \frac{1}{\sqrt{1 - \gamma^2 [\sin(k_x) - \sin(k_y)]^2}} dk_x dk_y \quad (48b)$$

For a physically acceptable, fully real spectrum one requires $|\gamma| \leq 1/2$. A numerical evaluation of (48b) shows that the maximal value of $\kappa + 1$ is

$$\kappa + 1 \Big|_{|\gamma|=\frac{1}{2}} = 1.39320388,$$

attained precisely on the line $\mu = \pm 2A_1$, with $\gamma = \pm 1/2$. Inserting this into (48a) yields

$$\mp \frac{4\mu}{J_1} = 2.20460520268 \quad \implies \quad \mu = \mp 0.5511513 J_1$$

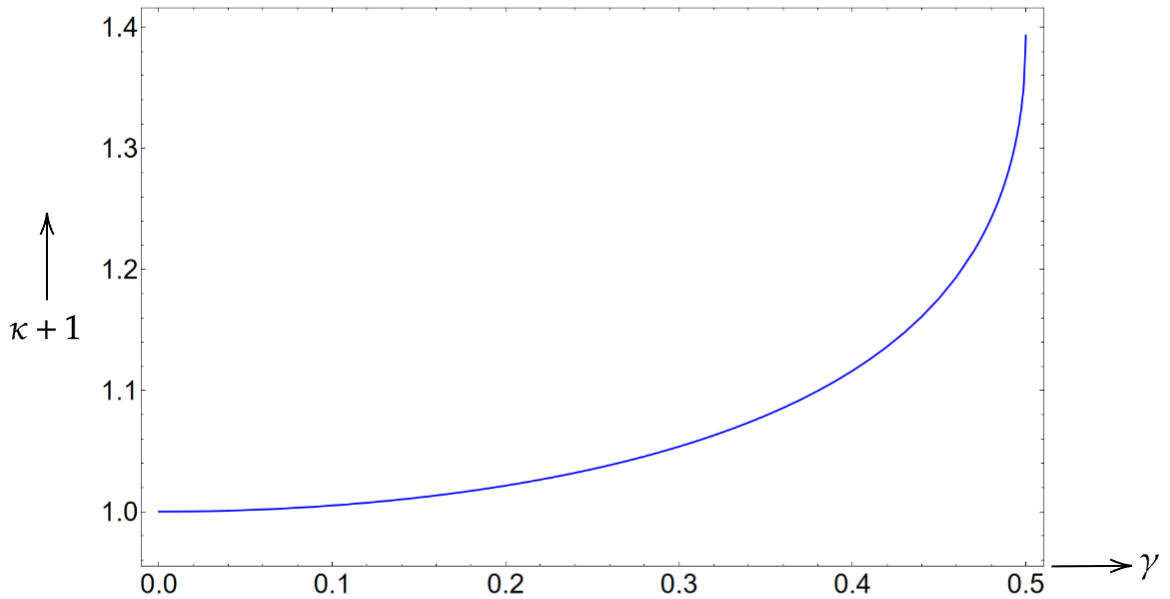


Figure 3: $\kappa + 1$ as a function of γ , when $J_2 = 0$ and any value of J_1 .

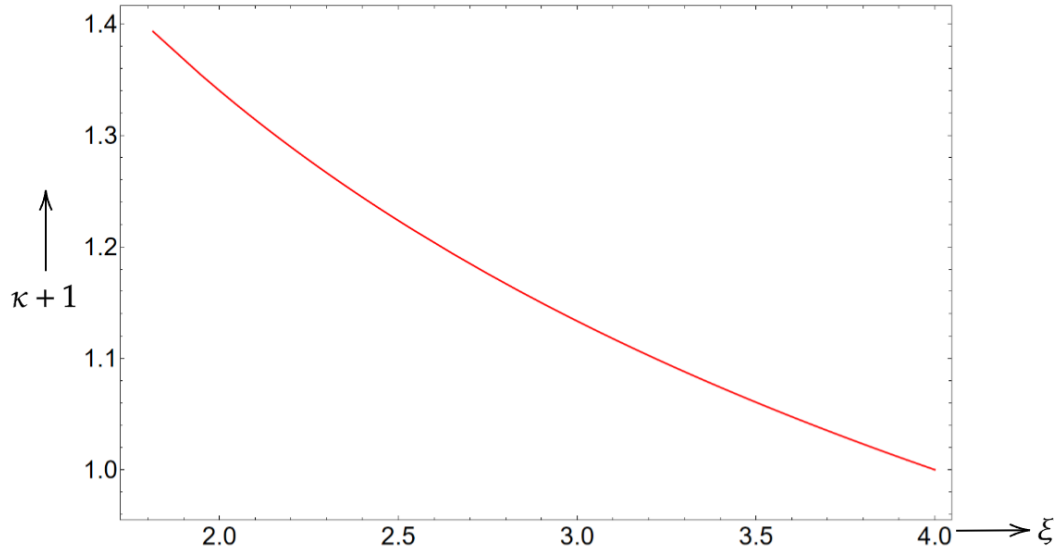


Figure 4: $\kappa + 1$ as a function of ξ , when $J_2 = 0$.

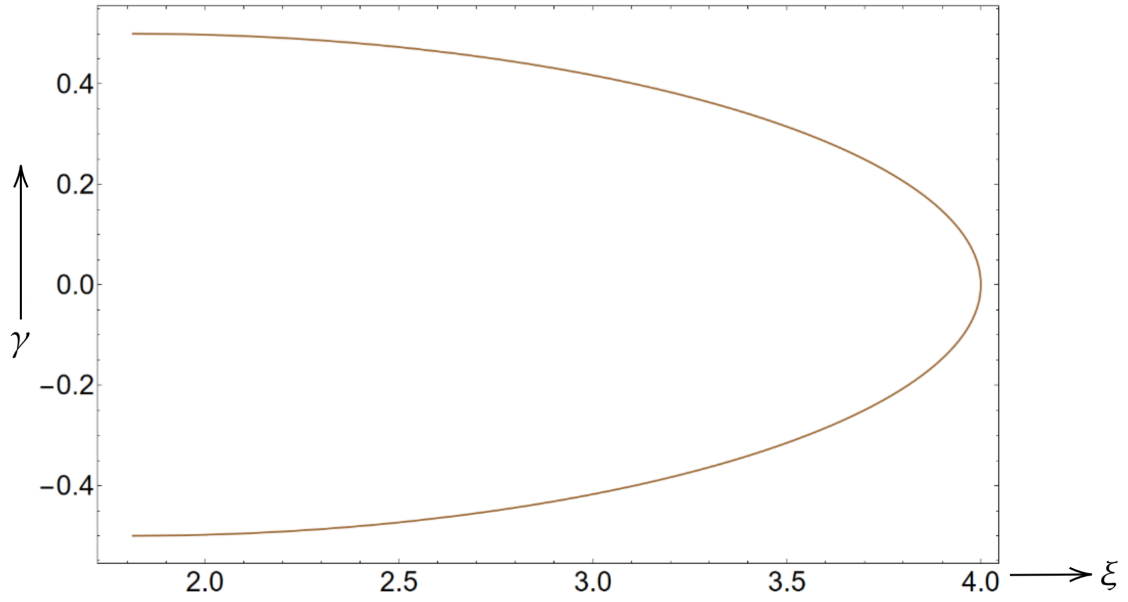


Figure 5: γ vs ξ , when $J_2 = 0$.

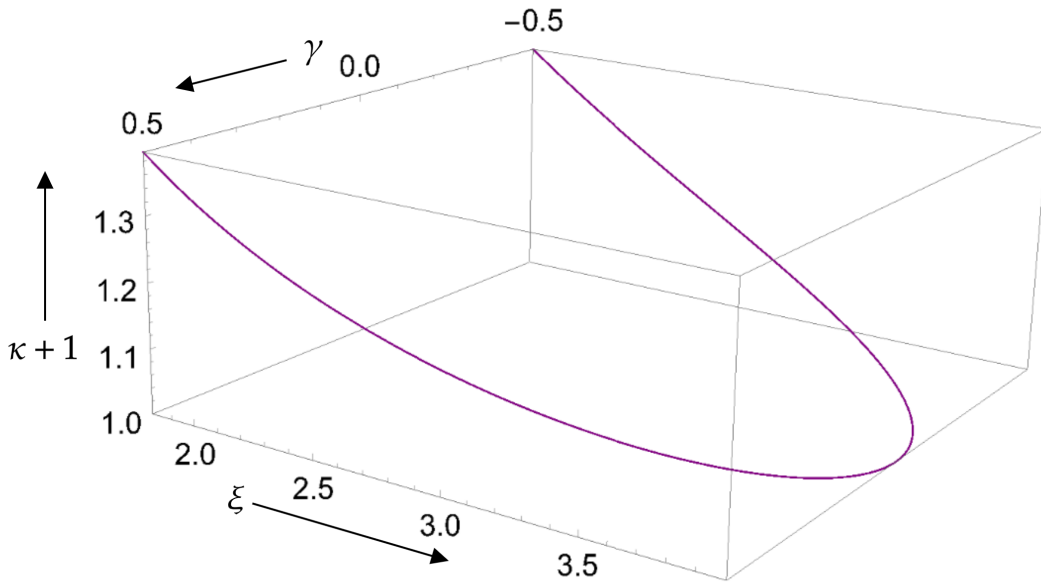


Figure 6: 3D plot of allowed values of ξ , γ and $\kappa + 1$.

5.3 Fixed point iteration for only B2.

In the case where both pairing and hopping amplitudes are present, i.e., $A_1, B_2 \neq 0$, the term A_1 cancels out from both sides of equation (47a). As a result, only equation (47b) remains as a genuine self-consistency condition. The remaining equations, (47a) and (47c), are then the root-finding conditions that must be satisfied simultaneously. The algorithm implemented is:

1. Fix a trial pair (A_1, μ) , and an initial guess value $B_2^{(0)}$, which guarantees a real spectrum $\omega(\vec{k})$ over the first Brillouin zone.
2. Solve (47b) by fixed-point iteration

$$B_2^{n+1} = f_{B_2}(B_2^n) \quad \text{until} \quad B_2^{(n)} \rightarrow B_2^* \quad \text{such that} \quad B_2^* = f_{B_2}(B_2^*)$$

3. The set $\{A_1, B_2^*, \mu\}$ must satisfy the below equation, if not then the pair (A_1, μ) is invalid.

$$1 = \frac{1}{(2\pi)^2} \iint_{\text{B.Z.}} \frac{J_1 [\sin(k_x) - \sin(k_y)]^2}{4 \sqrt{[\mu + 2B_2^* \sin(k_x) \sin(k_y)]^2 - A_1^2 [\sin(k_x) - \sin(k_y)]^2}} dk_x dk_y$$

Then, compute $\kappa + 1$ from (47c).

Results

This method produced physically meaningful solutions satisfying all three self-consistency conditions simultaneously. This solution lies on the plane $\mu = -2A_1$ in the three-dimensional parameter space (A_1, B_2, μ) . For $J_1 = J_2 = 1$, it was found that,

$$A_1 = 0.1482159, \quad B_2 = 0.0380264, \quad \mu = -0.2964319, \quad \implies \quad \kappa + 1 = 1.1501812$$

which solves (47a), (47b) and (47c) to seven decimal places. Moreover, when J_1 and J_2 are both scaled by the same factor (so that J_2/J_1 remains fixed), the triplet (A_1, B_2, μ) simply rescales in direct proportion, while $\kappa + 1$ remains unchanged. For instance, for $J_1 = J_2 = 2$ the solution becomes $A_1 = 0.2964319$, $B_2 = 0.0760529$, $\mu = -0.5928638$, again satisfying all three self-consistent equations and yielding the same $\kappa + 1 = 1.1501812$. This invariance suggests that, for fixed J_2/J_1 , the value of $\kappa + 1$ is universal.

For different values of J_2/J_1 , consider for instance $J_1 = 2$, $J_2 = 1$. The parameter values $A_1 = 0.3331432$, $B_2 = 0.05050237$, and $\mu = -0.66628647$ yield $\kappa + 1 \approx 1.1819259$, satisfying all three self-consistent equations simultaneously to seven decimal places. Even here, if J_1 and J_2 are rescaled by the same factor, the parameters are scaled with the same proportion. Notice that the value of $\kappa + 1$ has increased when J_2/J_1 has decreased to 0.5

To examine how $\kappa + 1$ depends on the ratio J_2/J_1 , I solved the self-consistent equations for various values of J_2/J_1 . The resulting data are summarized in the following table, listing some of the zero-flux $\mathbb{Z}_2$ spin liquid states that can exist for different choices of J_2/J_1 . All these solutions exist on the plane $\mu = -2A_1$.

J_1	J_2	J_2/J_1	A_1	B_2	μ	$\kappa + 1$
J_1	0	0	$0.27557565 J_1$	0	$-0.5511513 J_1$	1.3932038
40	7	0.175	7.7555	0.4984	-15.511	1.2323
3	1	0.333	0.531985	0.058362	-1.063971	1.201405
2	1	0.5	0.3331432	0.0505023	-0.66628647	1.1819259
1	1	1	0.1482159	0.0380264	-0.2964319	1.1501812
2	3	1.5	0.2757258	0.0945467	-0.5514517	1.1333461
1	2	2	0.1308566	0.0546172	-0.24337949	1.1225195
1	3	3	0.121689	0.065955	-0.243379	1.109257

Table 2: Solutions of the self-consistent equations for various values of J_2/J_1 .

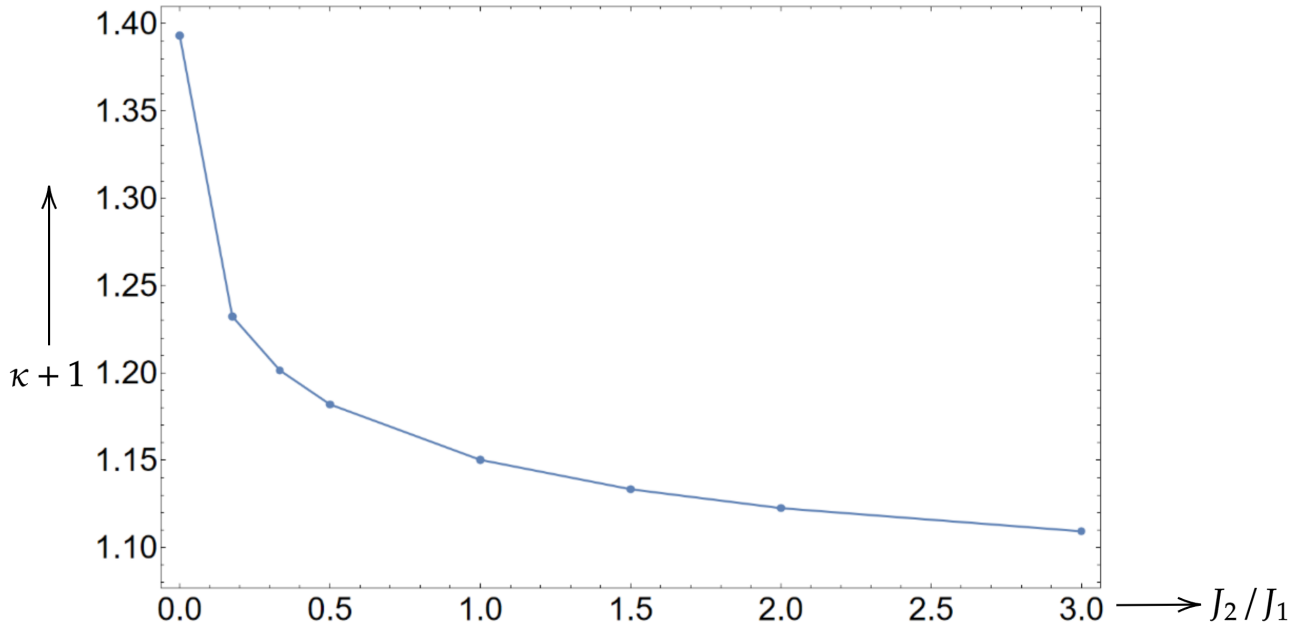


Figure 7: Graph of $\kappa + 1$ vs J_2/J_1 . As the ratio J_2/J_1 increases, the value of $\kappa + 1$ decreases.

6 Conclusion

The numerical study of the $\mathbb{Z}_2[0, 0]$ zero-flux mean-field equations focused on solving the three coupled integrals (47a)–(47c) for the variational parameters (A_1, B_2, μ) under the constraint $\kappa > 0$. Two primary algorithms were employed:

- **Fixed-Point Iteration for A_1, B_2 :** Equations (47a) and (47b) were iterated,

$$A_1^{(n+1)} = f_{A_1}(A_1^{(n)}), \quad B_2^{(n+1)} = f_{B_2}(B_2^{(n)}),$$

until convergence to fixed points (A_1^*, B_2^*) . Three types of stationary solutions emerged: a trivial paramagnet $(0, 0)$, a spurious pairing-free state $(0, B_2^*)$, and a nonphysical purely imaginary pairing. No valid real solution with both $A_1, B_2 \neq 0$ was obtained by this method alone.

- **Fixed-Point Iteration only for B_2 :** Holding trial values (A_1, μ) , the hopping amplitude was updated via

$$B_2^{(n+1)} = f_{B_2}(B_2^{(n)}),$$

with the remaining equations (47a) and (47c) enforced as root-finding conditions. This yielded a unique physically acceptable solution on the plane $\mu = -2A_1$. The solutions for A_1 , B_2 , and μ were found for various choices of J_2/J_1 .

Key observations across parameter space:

1. **Rescaling Invariance:** When both J_1, J_2 are multiplied by the same factor (fixing J_2/J_1), the solution (A_1, B_2, μ) rescales proportionally, while $\kappa + 1$ remains invariant.
2. **Dependence on J_2/J_1 :** As J_2/J_1 increases, the value of $\kappa + 1$ decreases monotonically.
3. **Hopping-Free Limit ($J_2 = 0$):** In the absence of B_2 , the problem reduces to two integrals in $\xi = -J_1/\mu$ and $\gamma = -A_1/\mu$, yielding a maximum $\kappa + 1 = 1.39320388$ at $|\gamma| = 1/2$ and $\mu = \mp 0.5511513 J_1$.

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